A Practical Guide to Autoregressive model

TOPIC -- AUTOREGRESSIVE MODEL

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Explicit mean/difference form of AR(1) process The AR(1) model is the discrete-time analogy of the continuous Ornstein-Uhlenbeck process. It is therefore sometimes useful to understand the properties of the AR(1) model cast in an equivalent form. In this form, the AR(1) model, with process parameter θ ∈ R {\displaystyle \theta \in \mathbb {R} } , is given by

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It can be seen that the autocovariance function decays with a decay time (also called time constant) of τ = 1 / (1 − ϕ) {\displaystyle \tau =1/(1-\varphi)} . The spectral density function is the Fourier transform of the autocovariance function. In discrete terms this will be the discrete-time Fourier transform:

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where B is the backshift operator, where ϕ (⋅) {\displaystyle \varphi (\cdot)} is the function defining the autoregression, and where ϕ k {\displaystyle \varphi _{k}} are the coefficients in the autoregression. The formula is valid only if all the roots have multiplicity 1. The autocorrelation function of an AR(p) process is a sum of decaying exponentials.

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which can be solved for all { ϕ m ; m = 1 , 2 , … , p } . {\displaystyle \{\varphi _{m};m=1,2,\dots ,p\}.} The remaining equation for m = 0 is

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Intertemporal effect of shocks In an AR process, a one-time shock affects values of the evolving variable infinitely far into the future. For example, consider the AR(1) model X t = ϕ 1 X t − 1 + ε t {\displaystyle X_{t}=\varphi _{1}X_{t-1}+\varepsilon _{t}} . A non-zero value for ε t {\displaystyle \varepsilon _{t}} at say time t=1 affects X 1 {\displaystyle X_{1}} by the amount ε 1 {\displaystyle \varepsilon _{1}} . Then by the AR equation for X 2 {\displaystyle X_{2}} in terms of X 1 {\displaystyle X_{1}} , this affects X 2 {\displaystyle X_{2}} by the amount ϕ 1 ε 1 {\displaystyle \varphi _{1}\varepsilon _{1}} . Then by the AR equation for X 3 {\displaystyle X_{3}} in terms of X 2 {\displaystyle X_{2}} , this affects X 3 {\displaystyle X_{3}} by the amount ϕ 1 2 ε 1 {\displaystyle \varphi _{1}^{2}\varepsilon _{1}} . Continuing this process shows that the effect of ε 1 {\displaystyle \varepsilon _{1}} never ends, although if the process is stationary then the effect diminishes toward zero in the limit. Because each shock affects X values infinitely far into the future from when they occur, any given value Xt is affected by shocks occurring infinitely far into the past. This can also be seen by rewriting the autoregression

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S (f) = σ Z 2 | 1 − ϕ 1 e − 2 π i f | 2 = σ Z 2 1 + ϕ 1 2 − 2 ϕ 1 cos ■ 2 π f {\displaystyle S(f)={\frac {\sigma _{Z}^{2}}{|1-\varphi _{1}e^{-2\pi if}|^{2}}}={\frac {\sigma _{Z}^{2}}{1+\varphi _{1}^{2}-2\varphi _{1}\cos 2\pi f}}}

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When ϕ 1 < 0 {\displaystyle \varphi _{1}<0} it acts as a high-pass filter on the white noise with a spectral peak at f = 1 / 2 {\displaystyle f=1/2} . The process is non-stationary when the poles are on or outside the unit circle, or equivalently when the characteristic roots are on or inside the unit circle. The process is stable when the poles are strictly within the unit circle (roots strictly outside the unit circle), or equivalently when the coefficients are in the triangle − 1 ≤ ϕ 2 ≤ 1 − | ϕ 1 | {\displaystyle -1\leq \varphi _{2}\leq 1-|\varphi _{1}|} . The full PSD function can be expressed in real form as:

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The simplest AR process is AR(0), which has no dependence between the terms. Only the error/innovation/noise term contributes to the output of the process, so in the figure, AR(0) corresponds to white noise. For an AR(1) process with a positive ϕ {\displaystyle \varphi } , only the previous term in the process and the noise term contribute to the output. If ϕ {\displaystyle \varphi } is close to 0, then the process still looks like white noise, but as ϕ {\displaystyle \varphi } approaches 1, the output gets a larger contribution from the previous term relative to the noise. This results in a "smoothing" or integration of the output, similar to a low pass filter. For an AR(2) process, the previous two terms and the noise term contribute to the output. If both ϕ 1 {\displaystyle \varphi _{1}} and ϕ 2 {\displaystyle \varphi _{2}} are positive, the output will resemble a low pass filter, with the high frequency part of the noise decreased. If ϕ 1 {\displaystyle \varphi _{1}} is positive while ϕ 2 {\displaystyle \varphi _{2}} is negative, then the process favors changes in sign between terms of the process. The output oscillates. This can be linked to edge detection or detection of change in direction.

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The terms involving square roots are all real in the case of complex poles since they exist only when $\varphi_2 < 0$ {\displaystyle \varphi _{2}<0} . Otherwise the process has real roots, and:

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Implementations in statistics packages R – the stats package includes ar function; the astsa package includes sarima function to fit various models including AR. MATLAB – the Econometrics Toolbox and System Identification Toolbox include AR models. MATLAB and Octave – the TSA toolbox contains several estimation functions for uni-variate, multivariate, and adaptive AR models. PyMC3 – the Bayesian statistics and probabilistic programming framework supports AR modes with p lags. bayesloop – supports parameter inference and model selection for the AR-1 process with time-varying parameters. Python – statsmodels.org hosts an AR model.

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This expression is periodic due to the discrete nature of the X_j {\displaystyle X_{j}} , which is manifested as the cosine term in the denominator. If we assume that the sampling time ($\Delta t = 1$ {\displaystyle \Delta t=1}) is much smaller than the decay time (τ {\displaystyle \tau }), then we can use a continuum approximation to B_n {\displaystyle B_{n}} :

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$S(f) = \frac{\sigma_Z^2}{1 + \varphi_1^2 + \varphi_2^2 - 2\varphi_1(1 - \varphi_2)\cos(2\pi f) - 2\varphi_2\cos(4\pi f)}$ {\displaystyle S(f)={\frac {\sigma _{Z}^{2}}{1+\varphi _{1}^{2}+\varphi _{2}^{2}-2\varphi _{1}(1-\varphi _{2})\cos(2\pi f)-2\varphi _{2}\cos(4\pi f)}}}